ThoughtAI

Value Prop: ThoughtAI for critical thinking because critical thinking is critical

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Problem Overview

People across many backgrounds increasingly view AI with skepticism. Many AI tools provide shortcuts that discourage critical thinking and replace human effort rather than supporting it. As companies adopt AI for speed and efficiency, tools often prioritize output over understanding. This has led to concern that human reasoning and problem solving are being devalued.

In education, this tension is especially visible. Students use AI to bypass reasoning, receive full solutions instantly, and disengage from the learning process. Our GR5 testing showed that while students want help, they do not want to lose control of their thinking. The challenge is designing AI that supports learning without replacing human reasoning.

Solution Overview

Solution: Instead of providing answers on demand, the AI withholds full solutions unless the student demonstrates active thinking.Progress is earned through visible reasoning, partial work, and engagement with guided prompts. The AI responds with hints, questions, and feedback rather than final answers, reinforcing learning without replacing human thought. This approach reframes AI as a support for reasoning rather than a shortcut.

Product: ThoughtAI

ThoughtAI is an iPad based learning app that applies this solution to handwritten problem solving.

Students write directly on the screen, import problems, or take photos of their work. ThoughtAI analyzes their reasoning and delivers contextual guidance, requiring students to complete each reasoning step before progressing or requesting further help.

By integrating AI into a student’s natural workflow, ThoughtAI supports reflection and understanding while preventing over reliance on AI generated answers.

Value Proposition

The value proposition of ThoughtAI is to help students learn through guided, step by step problem solving that reinforces critical thinking instead of shortcuts. By integrating AI directly into handwritten workflows, ThoughtAI keeps students in control of their reasoning, builds confidence in their understanding, and reframes AI as a learning partner rather than a replacement for human thought.

Our Task: Simple, Moderate, Complex

Simple: writing down the problem

This task demonstrates the most basic workflow, where a student opens the app, writes a problem, and explicitly submits their work using the “Check Work” action. The flow includes a clear submission step and a visible feedback state, addressing earlier confusion about when feedback is triggered. It is simple because it mirrors existing study behavior and requires minimal navigation or interpretation.

Moderate: taking picture of the problem or importing the problem

This task involves selecting an input method, capturing or uploading a problem, confirming the import, and transitioning into step by step problem solving. The inclusion of retake and confirm screens makes the system state explicit and ensures users understand when the problem is ready to be worked on. It is moderately complex because it introduces multiple decision points and system responses while remaining part of a common workflow.

Complex: AI voice guides them through like a tutor

This task represents a guided tutoring interaction where the AI controls progression by prompting the student to explain their reasoning before unlocking the next step. Students must respond verbally or with text, and confirmation feedback clearly indicates when a step is completed correctly. It is complex because it requires sustained cognitive effort, interpretation of prompts, and enforced reasoning before progression.

Usability goals and Key Measurements

GOAL & MEASUREMENT # 1

Clear Learning Flow

Students should easily understand how to open the right notebook, ask a question, receive help, and continue working.

Measurement

Navigation errors on repeated attempts

If students make fewer mistakes when repeating the same flow (opening a notebook, asking a question, continuing work), this suggests the learning flow is clear.

Why

A clear learning flow helps students focus on learning instead of figuring out how to use the app.

GOAL & MEASUREMENT # 2

AI That Helps Students Keep Going

AI should support students when they are stuck in a way that encourages progress rather than giving answers.

Measurement
AI engagement frequency during a problem

We observe how many times a student clicks the AI button while working on a single problem and whether AI use decreases as they make progress.

Why
This shows whether AI supports persistence and learning rather than replacing student thinking.

Progress Towards These Goals

Clear Learning Flow

- Earlier designs made it unclear how students should move from choosing a notebook to asking a question and continuing work. The revised design adds clearer entry points and confirmation cues so students understand how the learning flow works.

AI That Helps Students Keep Going

- Earlier AI feedback appeared without clear control. The revised design presents AI guidance as step based help with a clear option to close it, allowing students to apply guidance and keep working.

How the revised design supports these goals

Clear Learning Flow

The revised design helps students move smoothly from opening the correct notebook to asking a question and continuing their work. The added search feature allows students to quickly find the notebook that matches their subject, reducing confusion at the start. Clear entry points and confirmation cues make it easier to understand how the learning flow works without extra explanation.

AI That Helps Students Keep Going

AI guidance is now presented as optional, step based help with a clear way to close it once the student is done. This gives students control over when they receive help and prevents the AI from taking over the problem. As a result, students can use AI support to move forward when stuck and then continue working on their own.

Necessary Future Improvements

Clear Learning Flow

- Ensure future features follow the same open → ask → get help → continue pattern so students always know how to move forward.

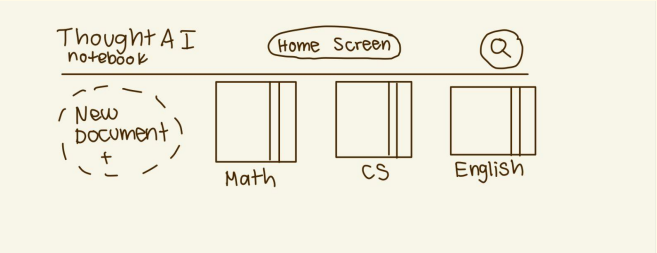
AI That Helps Students Keep Going

- Further refine AI responses to sound more engaging and inspirational to encourage students to utilize step by step problem solving with AI rather than not relying on it at all or seeking just seeking the full solution

Revised Interface Sketches

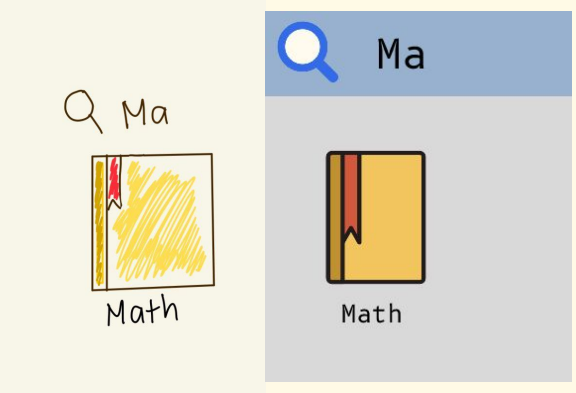
Change #1

Search to Find the Right Notebook



Previously

The home screen displayed a search bar, but it was not connected to any clear action. Students still had to visually scan notebooks and guess which one to open, making it unclear how to begin when multiple notebooks were present.



Now

The search bar allows students to directly search for and select the notebook they want to use. This makes it easier to quickly open the correct subject notebook and start asking questions without unnecessary steps.

Figma: There is an actual page now that is dedicated to the search tool to make it easier for students to find a notebook because they might create many notebooks and might have difficulty finding the one they are searching for.

Benefits

Clear Learning Flow

This change helps students move smoothly from opening the app to selecting the correct notebook. By allowing students to search by subject, the revised design makes it clearer how to begin working and continue learning without confusion.

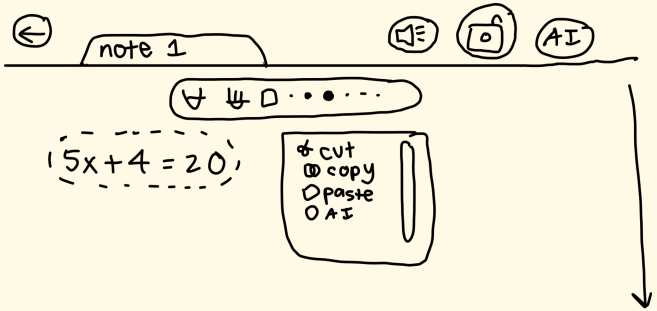
Explanation of Change #1

During low fi testing and studio feedback, students expressed uncertainty about how to begin when multiple notebooks were available. Although a search bar was present on the home screen, it did not clearly support notebook selection, causing users to pause and visually scan the page before choosing where to start. This hesitation indicated that the starting point of the learning flow was unclear.

In response, the revised design connects the search bar directly to notebook selection, allowing students to search by subject and open the appropriate notebook immediately. This change addresses the observed hesitation by making the first step more obvious and reducing unnecessary navigation. By helping students quickly find the correct notebook, the revised design better supports user needs for a clear and efficient starting point.

Change #2

Giving Students Control Over AI Help

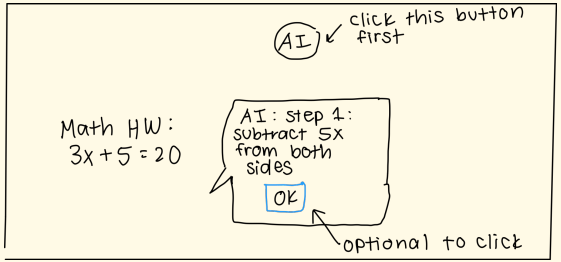


Previously

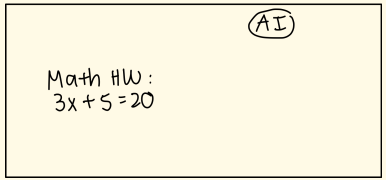
Once students activated the AI assistant, it remained on the screen without a clear way to dismiss it. This meant AI guidance was always visible, even when students felt they understood the next step or wanted to continue working on their own..

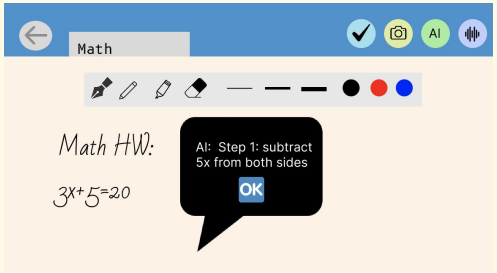
Now

AI guidance appears only when students choose to use it and includes an OK button that allows them to close it when they are done. Students can decide when they want help and when they want to continue working without AI on the screen.

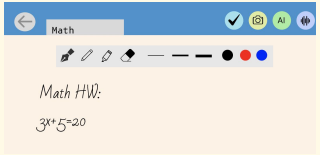


If student clicked OK
AI box dissappears





Figma Model



After Clicking OK

Benefits

AI That Helps Students Keep Going

By making AI optional and dismissible, students can use guidance as support rather than feeling dependent on it. This encourages students to apply the hint and continue working instead of waiting for the AI to guide every step.

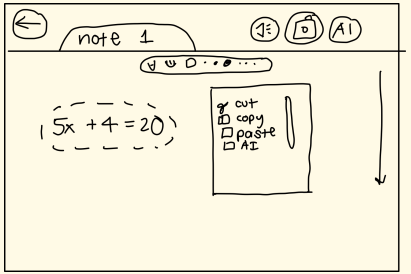
Explanation of Change #2

During low fi testing and studio feedback, students noted that constant AI presence felt distracting and made it harder to focus once they understood what to do next. Some users expressed that they wanted help only when they were stuck, not throughout the entire problem.

In response, the revised design gives students control over AI visibility by adding an OK button to close guidance and making AI something students engage with intentionally. This change directly addresses user needs by supporting focus, reducing unnecessary distraction, and allowing AI to act as help when needed rather than a constant presence.

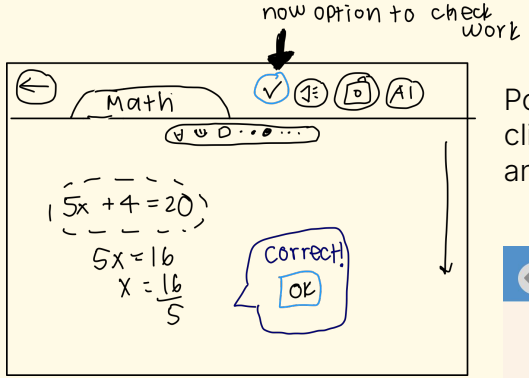
Change #3

Letting Students Check Their Work Independently



Previously

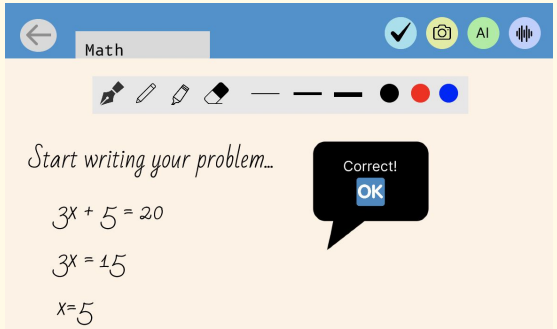
Students relied entirely on the AI assistant to know whether their work was correct. There was no built in way to check an answer without asking AI for help, even when students wanted to work independently.



Now

Students can use a check button to see whether their work is correct or incorrect. The check provides simple feedback without hints or explanations, allowing students to verify their understanding while continuing to solve the problem on their own.

Pop-up will disappear after clicking ok and will say incorrect if answer is incorrect



Figma

Benefits

Clear Learning Flow

The check button helps students move forward with confidence by confirming whether their current work is correct before continuing. This makes it clearer when to keep working and when to revisit a step.

AI That Helps Students Keep Going

By offering correctness feedback without hints, this feature reduces over reliance on AI. Students can validate their work independently and turn to AI only when they truly need guidance.

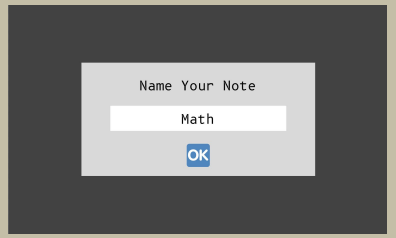
Explanation of Change #3

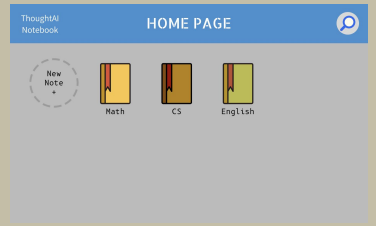
During low fi testing and studio feedback, students expressed that they wanted a way to confirm their answers without immediately involving AI. Relying solely on AI made some users feel dependent on assistance even when they were confident in their reasoning.

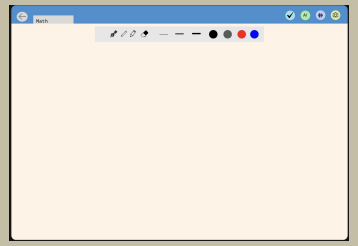
In response, the revised design introduces a check button that gives clear correctness feedback without providing solutions or hints. This addresses user needs by supporting independent problem solving while still allowing AI to be used as optional support when students are stuck.

Medium-fi task flows

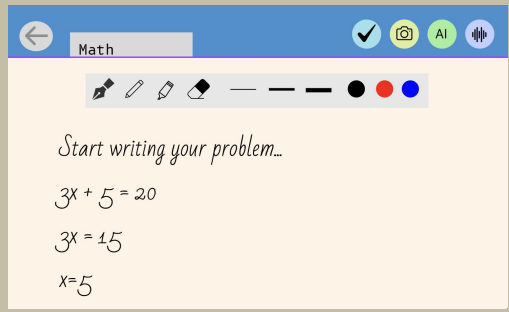
Simple Writing down the problem

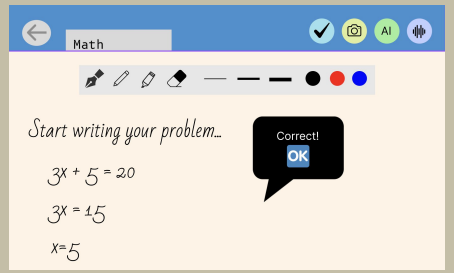




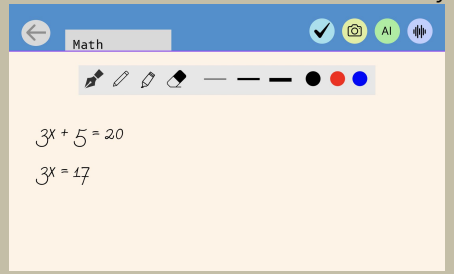


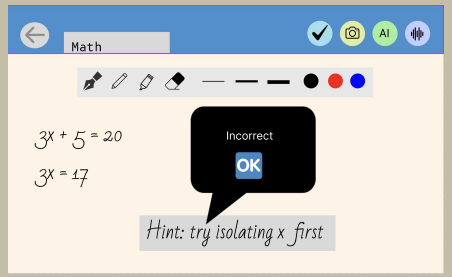
Case 1: the user solve it correctly





Case 2: the user solve it incorrectly

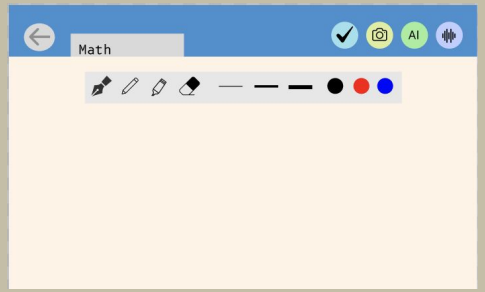




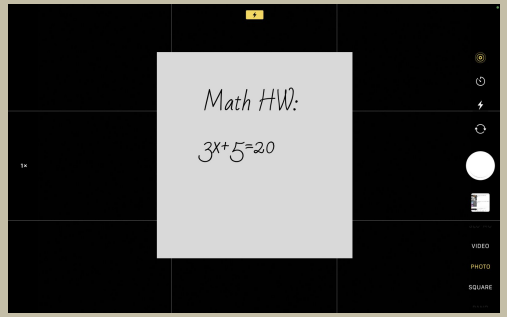
Moderate

Taking picture of the problem or importing the problem

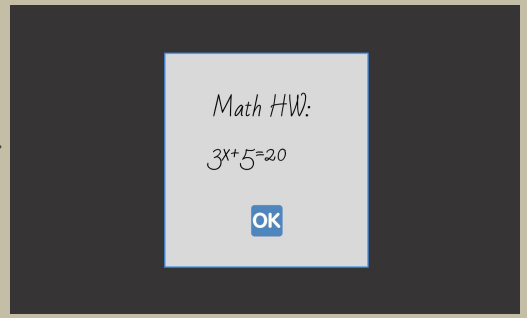
Click New Note



Taking picture of the problem

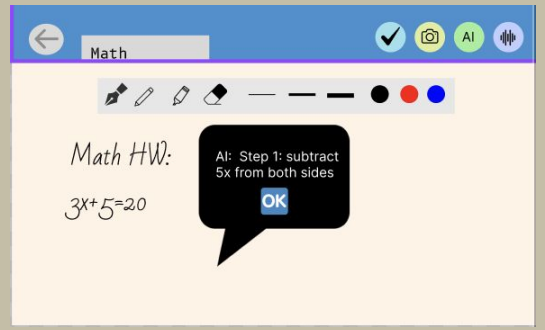


AI identifies the problem



Start doing the problem

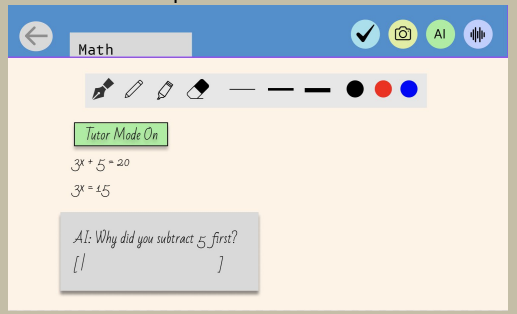
AI give hints for the problem



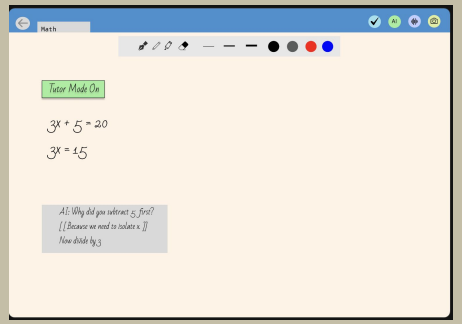
Complex AI voice guides them through like a tutor

Turn AI Mode on

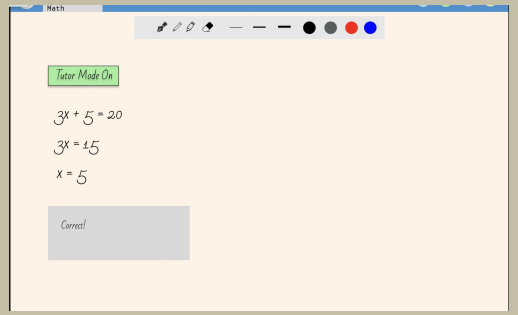
AI ask a question



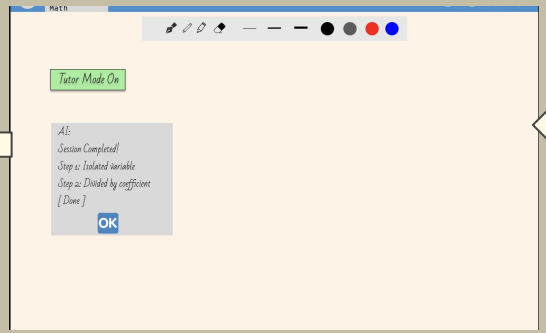
User respond to the question and AI give hints for next step



AI grading the problem



AI is done



Prototype Implementation

Why the pros/cons tradeoff was acceptable for your implementation?

Our implementation intentionally prioritizes student control and independent problem solving over automation and constant guidance. While adding features such as a dismissible AI assistant and a check button introduces extra decisions for the user, this tradeoff was acceptable because it better supports how students actually work through problems. One tradeoff of allowing students to close the AI assistant is that some users may choose to disengage from AI help too early. However, this was an intentional decision to reduce distraction and over reliance on AI. Giving students control over when AI appears allows them to focus when they already understand a step and seek help only when they are truly stuck.

Similarly, introducing a check button that only confirms correctness without giving hints means students may still need to reason through mistakes on their own. While this provides less immediate guidance than AI, it encourages independent thinking and reduces dependency. Students can validate their understanding quickly and decide whether they need further help, rather than defaulting to AI for confirmation.

Finally, connecting the search bar to notebook selection adds a small amount of interface complexity, but this was an acceptable tradeoff to make the starting point of the app clearer. Helping students quickly find the correct notebook reduces confusion at the beginning of a session and supports a smoother learning flow overall. Overall, these tradeoffs were acceptable because they align with the core goal of ThoughtAI as a learning support tool rather than a solution provider. The revised design balances guidance with independence, helping students stay engaged and in control while solving problems.

Limitations of the prototype: What was left out? Why?

No real AI processing: The prototype simulates AI responses and reasoning checks, but does not implement actual evaluation logic or natural language understanding. This was left out because the goal was to demonstrate interaction design, not backend feasibility.

Simplified voice interaction: Speech recognition and transcription are represented conceptually but we did not model accuracy issues, mishearing, or correction flows to keep the prototype focused and manageable.

Limited problem scope: We only demonstrated a simple algebra example rather than multiple problem types or longer multi-step equations. This was intentional to highlight workflow differences rather than mathematical complexity.

Minimal edge-case handling: We did not include ambiguous handwriting, partially correct answers, or system errors. These were excluded to avoid overcomplicating the core task comparison.

Hard Coded Features:

- The algebra problem (e.g., 3x + 5 = 20) is fixed and does not change.
- Correct and incorrect answer paths are predefined.
- Hints and feedback messages are static and not adaptive.
- The summary screen always displays the same completed steps.
- he system does not handle alternative solution methods or varied user inputs.

Wizard-of-Oz Features:

- AI feedback and reasoning evaluation are pre-written and triggered manually through prototype transitions.
- Tutor Mode appears to evaluate student explanations, but responses are scripted rather than dynamically generated.
- Step unlocking in Complex mode is simulated through predetermined screen transitions rather than real reasoning assessment.

c. What was the impact of these limitations on the user experience?

Reduced realism: Because AI evaluation and voice processing were simulated, users would not experience true adaptive feedback, which may limit how realistic the tutoring interaction feels.

Limited adaptability: Since problems and responses were hard-coded, the system cannot handle alternative solution methods or unexpected inputs, reducing flexibility.

Predictable interactions: Feedback and progression follow fixed paths, which may make the experience feel scripted rather than intelligent.

Constrained exploration: Users cannot test edge cases or different strategies, limiting the depth of interaction in Complex mode.

Maintained clarity of flow: Despite these limitations, the simplified implementation helps clearly communicate differences in task complexity (Simple vs. Moderate vs. Complex), which was the primary goal of the prototype.

Appendix

Figma Link

https://www.figma.com/proto/cyCgAJYGy612MWPMXerbW9/ThoughtAI?node-id=32-78&p=f&t=PRApe2aUjgMxNbF3-0&scaling=scale-down&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=32%3A77

Read Me Link

https://docs.google.com/document/d/1VAVI8iyTiMWJcufIeCyEmv0zfN8dpG8DJe1Fra4Yny0/edit?usp=sharing